4.2.1

AI25BTECH11038 – Tejas Uppala

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Question

Find the direction and normal vectors of the following line $2x + 3y = 9$

Solution

For a general line y = mx + c, it can be converted into its vector form in the following way,

$$y = mx + c \tag{0.1}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} x \\ mx + c \end{pmatrix} = \begin{pmatrix} 0 \\ c \end{pmatrix} + x \begin{pmatrix} 1 \\ m \end{pmatrix} \tag{0.2}$$

yielding,

$$\mathbf{x} = \mathbf{h} + k\mathbf{m} \tag{0.3}$$

where **h** is any point on the line and **m** is the direction vector of the line,

$$\mathbf{m} = \begin{pmatrix} 1 \\ m \end{pmatrix} \tag{0.4}$$

The given line can be written as,

$$y = \left(\frac{-2}{3}\right)x + \frac{9}{3} = \left(\frac{-2}{3}\right)x + 3 \tag{0.5}$$

on correlating,

$$\mathbf{m} = \begin{pmatrix} 1 \\ \frac{-2}{3} \end{pmatrix} or \begin{pmatrix} 3 \\ -2 \end{pmatrix}, \mathbf{h} = \begin{pmatrix} 0 \\ 3 \end{pmatrix} \tag{0.6}$$

The **normal vector** (n) of the line is $\begin{pmatrix} -m \\ 1 \end{pmatrix}$. Therefore,

$$\mathbf{n} = \begin{pmatrix} \frac{2}{3} \\ 1 \end{pmatrix} or \begin{pmatrix} 2 \\ 3 \end{pmatrix} \tag{0.7}$$